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Paleozoic history of the Kara microcontinent and its relation to Siberia and Baltica: Paleomagnetism, paleogeography and tectonics

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Abstract

The tectonic history of the Kara microcontinent on the northern part of the Taimyr-Severnaya Zemlya region, Siberian Arctic, has been the subject of considerable dispute, especially with respect to whether it was attached to Siberia or Baltica, or was an independent plate during the Late Neoproterozoic and Early Paleozoic. One way to resolve this controversy is by quantitative paleomagnetic data, and in this article, we present the results of a paleomagnetic study of Early Paleozoic strata from October Revolution Island in the Severnaya Zemlya Archipelago, which allow definition of the paleogeographic position of the Kara microcontinent during the Early Paleozoic. Paleomagnetic poles for three consecutive time intervals, i.e., Late Cambrian–Early Ordovician, Middle–Late Ordovician and Late Silurian, have been determined. These new data demonstrate that Kara was not part of Siberia or Baltica during the Early Paleozoic, but support models that it was an exotic microcontinent with an independent tectonic history. The Paleozoic tectonic evolution of the Kara microcontinent was dominated by strike–slip displacements associated with a transform margin that resulted in collision with the Siberian craton in the Late Carboniferous to Permian. This interpretation provides a new foundation to explain the Paleozoic history of the Kara microcontinent.
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Keywords: Siberian Arctic; Taimyr–Severnaya Zemlya region; Kara microcontinent; Paleozoic; Paleomagnetic constraints; Tectonic evolution

1. Introduction

In recent years, most researchers have considered the Kara plate or microcontinent, exposed in the northern part of the Taimyr fold belt in the Taimyr Peninsula and the Severnaya Zemlya Archipelago, to be exotic to the Siberian craton and to have been accreted to it during the Late Paleozoic (Zonenshain et al., 1990; Uflyand et al., 1991; Vernikovsky et al., 1995; Bogdanov et al., 1998). Structural, geological, geophysical, petrological–geochemical and geochro-

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nological data support this interpretation, but a lack of quantitative paleomagnetic information has not allowed an estimate of the paleogeographic position of the Kara microcontinent in the Early Paleozoic or of the detailed kinematics of its subsequent collision with Siberia. The time of collision and the tectonic character of the interaction between Siberia and Kara are debated (Zonenshain et al., 1990; Vernikovsky, 1996, 1997; Lawver et al., 2002). All existing tectonic models of the Kara–Siberia connection have been based on structural and petrologic information. The first paleomagnetic data for the region were reported by Metelkin et al. (2000), and in this contribution, we represent the paleomagnetic results with data from Paleozoic strata of the October Revolution Island (Severnaya Zemlya Archipelago), which allow improved paleo-reconstructions for the Kara microcontinent.

2. Geological setting

The Taimyr–Severnaya Zemlya fold-thrust belt is situated in Arctic Siberia along the Kara Sea coast between the Ural–Novaya Zemlya fold belt to the west and the Verkhoyansk fold belt to the east. It is subdivided into southern, central and northern domains separated by the major Pyasina–Faddey and Main Taimyr thrusts (Fig. 1). The Taimyr margin of the Siberian platform has undergone at least two stages of deformation, i.e., Late Neoproterozoic and Late Paleozoic–Early Mesozoic (Zonenshain et al., 1990; Vernikovsky, 1996, 1997; Inger et al., 1999).

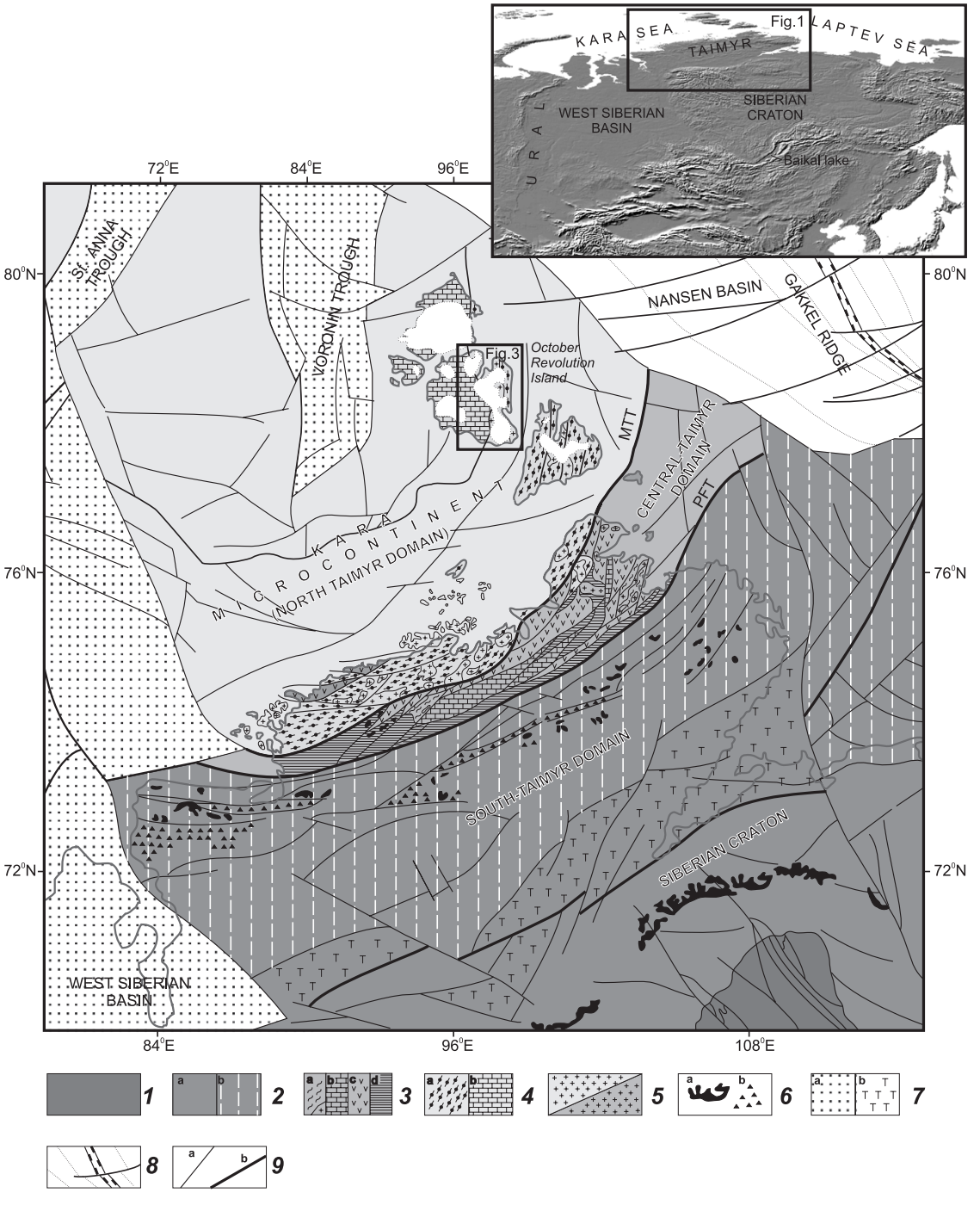
The South Taimyr Domain is composed of unmetamorphosed Ordovician to Permian deposits and represents the northwest-facing Paleozoic passive margin of the Siberian continent (Uflyand et al., 1991; Vernikovsky, 1996; Bogdanov et al., 1998; Inger et al., 1999). The section is largely represented by carbonate and siliciclastic shallow-sea strata. The Upper Permian–Lower Triassic part of the section is composed of volcanic rocks, including basalts, tuffs, dolerite sills and dikes that are linked to the Siberian traps. All strata, including the traps, are deformed by south-verging folds and thrusts, which decrease in intensity towards the south and are cut by post-tectonic A-type granite and syenite bodies that have yielded emplacement ages of 249–241 Ma (Vernikovsky et al., 2003).

The Central Taimyr Domain is mainly composed of Mesoproterozoic metamorphic rocks and Neoproterozoic sedimentary and volcanic rocks, including ophiolites and magmatic rocks with island arc affinity (Uflyand et al., 1991; Vernikovsky et al., 1996, 1998a; Pease et al., 2001; Vernikovsky and Vernikovskaya, 2001). These units were accreted to the Siberian margin (i.e., the South Taimyr Domain) in the Late Neoproterozoic (~600 Ma), and covered by Late Neoproterozoic (Vendian) and Paleozoic successions (Bezzubtsev et al., 1986; Zabiyaka et al., 1986; Uflyand et al., 1991; Vernikovsky, 1996).

The North Taimyr Domain (including the northern part of the Taimyr Peninsula and Severnaya Zemlya Archipelago) represents the Kara microcontinent. Continuity of geophysical data (aeromagnetic and gravity) suggests that structures in the basement of the North Taimyr Domain continue across much of the Kara shelf (Bogolepov et al., 1991; Bogdanov et al., 1998). The main structures in the Kara, including major dextral strike–slip faults, are sub-parallel to the arc of the collisional belt (Fig. 1). To the west, the plate is abruptly terminated by the N–S-trending St. Anna trough, which is expressed in the modern relief of the sea-floor as a neotectonic trench (Matishov et

Fig. 1. Tectonic scheme of the North Siberia and Kara Sea (after Vernikovsky, 1996; Bogdanov et al., 1998). Legend: 1–2–Siberian craton: 1–Archean and Paleoproterozoic basement; 2–Phanerozoic cover (a), including Paleozoic and Early Mesozoic sediments of the South Taimyr Domain (b); 3–Neoproterozoic accretionary belt (Central Taimyr Domain): metamorphic terranes with 920–850 Ma granite (a), passive continental margin terranes with Neoproterozoic carbonate deposits (b), island arc and back arc terranes, ophiolite with 700–630 Ma plagiogranite (c), Vendian to Early Carboniferous sedimentary cover (d); 4–Kara microcontinent (North Taimyr Domain): Neoproterozoic to Cambrian turbidites of the Kara continental slope (a), Paleozoic coastal-marine, lagoonal and continental sediments of the Kara cover (b); 5–collisional granite 300–265 Ma; 6–dolerites (a) and flood basalts with associated volcanic–sedimentary deposits (b) of the Permian–Triassic trap formation; 7–Meso-Cenozoic troughs, suboceanic crust areas (a), and continental rifts (b); 8–oceanic crust areas and rifts; 1–major faults (a), including domain boundary (b).

al., 1995). The southern boundary of the North Taimyr domain (i.e., of the Kara microcontinent) is marked by dextral strike–slip faults (Bogdanov et al., 1998), and the northern and northeast margins of the plate are truncated by the Nansen depression of the Arctic Ocean.



Two tectonostratigraphic zones can be distinguished in the North Taimyr domain:

(1) A Neoproterozoic to Cambrian turbidite succession, consisting of rhythmically alternating sandstone, siltstone and shale interpreted as continental slope sediments, which display evidence of greenschist to amphibolite facies of metamorphism (Vernikovsky, 1996);
(2) Unmetamorphosed Upper Cambrian to Devonian platformal strata comprising shallow-water lagoonal and coastal-marine facies (Makar'ev et al., 1982; Markovsky and Smirnova, 1982; Khapilin, 1982).

3. Sampling and instrumentation

We sampled representative sections from the Upper Cambrian to Silurian of the Kara succession (Figs. 2, 3). The Upper Cambrian–Lower Ordovician strata are exposed in the southern part of October Revolution Island, where they are gently folded. The upper part of the Upper Cambrian Kurchavinskaya Formation, which was sampled for paleomagnetic studies (Fig. 3), is represented by grey, graded, cross-bedded sandstone with abundant trace fossils, intercalated with dark-grey siltstone and dark claystone. Limestone concretions are rare. On the basis of rich fauna, the age of the upper part of the Kurchavinskaya Formation has been constrained to between the Agnostus–Peltura zones of the Upper Cambrian (Bogolepova et al., 2001). There is an unconformable contact between the Upper Cambrian Kurchavinskaya Formation and the overlying Kruzhilikhinskaya strata. The Lower Ordovician Kruzhilikhinskaya Formation is mainly composed of sandstones with conglomerate and gritstone lenses, and interbeds of siltstones, sandstones, acid to intermediate volcanic rocks, including effusives and tuffs, clayey dolomites and limestones with fauna. The macrofossils indicate an Arenig age for this formation (Markovsky and Makar'ev, 1982). In some areas the Cambrian and Ordovician formations comprise a monotonous sequence, which it has not proven possible to subdivide. However, the geological data permit correlation of these rocks with the Kurchavinskaya formations, and thus restrict their upper age limit as Tremadocian.

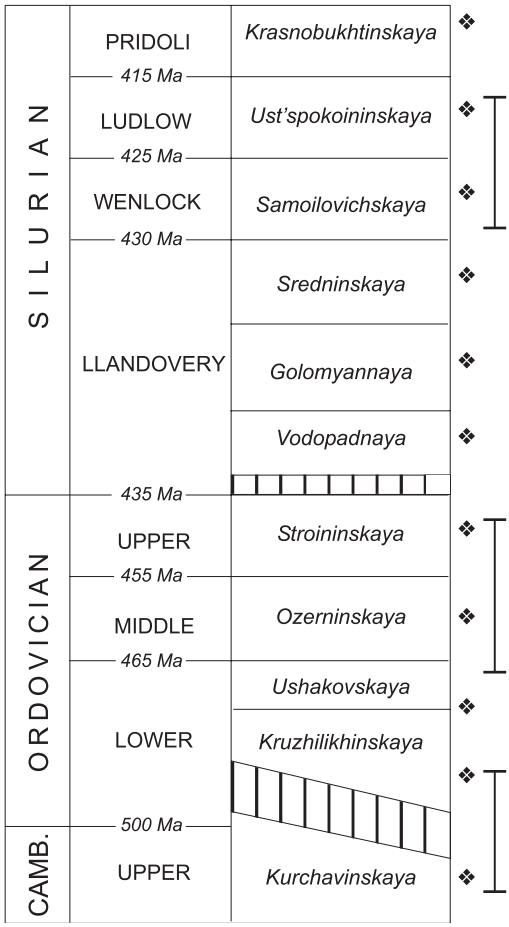


Fig. 2. The Upper Cambrian–Silurian stratigraphy of the October Revolution Island with points of paleontology sampling sites (diamonds) and paleomagnetic sampling intervals (snippets). Stripped parts in the column represent hiatus.

The Ordovician rocks were also sampled for both paleontological and paleomagnetic studies in the basin of the Ushakova River near Krasnaya Bay (Fig. 3). The Stroininskaya Formation is characterized by sandstone, dolostone and carbonaceous dolostone, marl and rare limestone, and an Ashgillian age has been suggested for this formation on a basis of macrofossils (Markovsky and Makar'ev, 1982). The underlying Ozerninskaya and Ushakovskaya Formations are composed of marl, variegated siltstone and sandstone, gypsum, and limestone lenses. The fossil assemblage includes stromatolites, algae filaments,

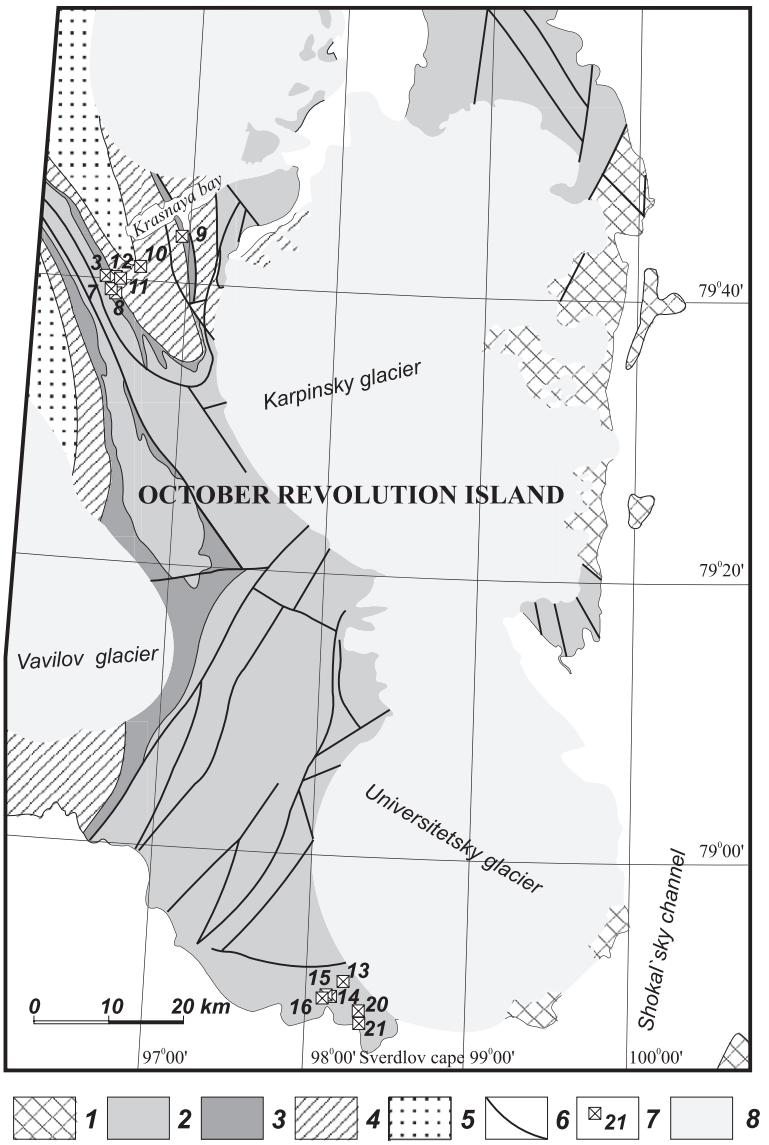


Fig. 3. Geological scheme of October Revolution Island. Legend: 1–5–geological units: 1–Neoproterozoic to Vendian; 2–Cambrian to Early Ordovician; 3–Ordovician; 4–Silurian; 5–Devonian; 6–faults; 7–paleomagnetic sampling sites and their numbers; 8–glaciers.

gastropods and brachiopods, which constrain the age of those formations from uppermost Early Ordovician to the Middle Ordovician, Arenig–Llanvirn (Markovsky and Makar'ev, 1982).

The Wenlock–Ludlow is the next stratigraphic level that has been investigated paleomagnetically (Figs. 2, 3). The Samoilovichskaya Formation has been studied in the upper reaches of the Ushakova River. It is represented by grey and dark-grey stromatolitic and oncolitic limestone intercalated with grey clay-rich dolomitic limestone. In the middle and upper parts of the formation, algal bioherms are common. Fossils are abundant and include crinoids, stromatoporoids, tabulate corals, ostracodes, gastro-

pods, brachiopods, conodonts and vertebrates, which have yielded a Wenlock age (Matukhin et al., 1999). The overlying Ust'spokoininskaya Formation is represented by sandy (rarely clayey) limestone, marl, dolostone and sandstone of Ludlow age (Matukhin et al., 1999).

Paleomagnetic sampling was carried out with a portable drill and both magnetic and sun compasses were used for sample orientation. The laboratory experiments were carried out at the Paleomagnetic Center of the Institute of Geology, Siberian Branch of the Russian Academy of Sciences (Novosibirsk, Russia) and the Paleomagnetic Laboratory at Lund University (Sweden). Magnetic susceptibility values were measured with a Burakov susceptibility meter (Novosibirsk). Direction and intensity of the natural remanent magnetization were measured with a JR-4 spinner magnetometer (Novosibirsk), a cryogenic HSM magnetometer (Novosibirsk), and a 2G Enterprise magnetometer (Lund), the latter instrument having a built-in alternating field demagnetizer. Stepwise thermal demagnetization, up to 700 °C in 30–40 °C steps, was carried out on a Schonsted TSD-2 furnace (Lund) and an Aparin shielded furnace system (Novosibirsk). All laboratory measurements were carried out using standard techniques, and control measurements on twin samples analyzed in both Novosibirsk and Lund yielded identical results. Using various techniques of paleomagnetic data interpretation, including principal component analysis (Kirschvink, 1980), remagnetization circles (Halls, 1976) and combined analysis of remagnetization circles and direct observations (McFadden and McElhinny, 1988), it was possible to determine site mean directions for all sites, even those with multiple components of NRM and several overlapping spectra of blocking temperature or coercivity.

4. Paleomagnetic results

4.1. Upper Silurian

NRM intensity is low, especially in calcareous samples (typically less than 1 mA/m), and the susceptibility generally does not exceed 30×10^{-6} SI units. Three tested Silurian sites showed different directional behavior during stepwise thermal and AF demagnetization.

NRM stability for sandstone samples (site 10) is high with respect to both AF and thermal demagnetization. Low-temperature components are practically absent and above 200 °C vector plots show univectorial decay towards the origin of the diagram. The NRM is unblocked at temperatures at about 680 °C (Fig. 4). AF demagnetization did not change the direction and intensity of NRM at 100 mT AF treatments, under which conditions the NRM intensity was approximately 95% of its initial value. This demonstrates the importance of high-coercivity hematite as the remanence carrier.

Thermal or AF demagnetization of limestone samples (site 11) did not isolate high-temperature or high-coercivity characteristic remanence components (ChRC) (i.e., did not yield linear segments on vector plots) and low-temperature components are scattered. NRM intensity was low below ca. 300 °C, but above 340–420 °C, the intensity increased sharply by a factor of about 10 followed by irregular decay curves and directional instability, indicating that the samples underwent thermochemical changes. Below these temperatures, the directional trends are confined to great-circle trends (Fig. 4). Stability to AF demagnetization is relatively high. Median destructive fields (MDF) vary from 10 to 32 mT. Low-coercivity components are identified between 12 and 28 mT, but they are scattered (sub-vertical in geographic coordinates). High-coercivity components were not isolated and demagnetization data are distributed along great circles (Fig. 4), testifying to overlapping coercivity spectra. Nevertheless, the intersections of calculated great circles based on individual thermal and AF demagnetization data (Fig. 4) allow estimation of a site mean direction that is nearly antipodal to ChRC identified from the sandstones (in stratigraphic coordinates).

NRM stability during thermal demagnetization is rather low for marl samples (site 12). Low-temperature components are observed below 230 °C and they are closely grouped (subvertical in geographic coordinates). The low-temperature component comprises ca. 90% of the total NRM. At higher temperatures, directions are displaced along great circle trends and indicate the presence of high-temperature components (Fig. 4). However, high-

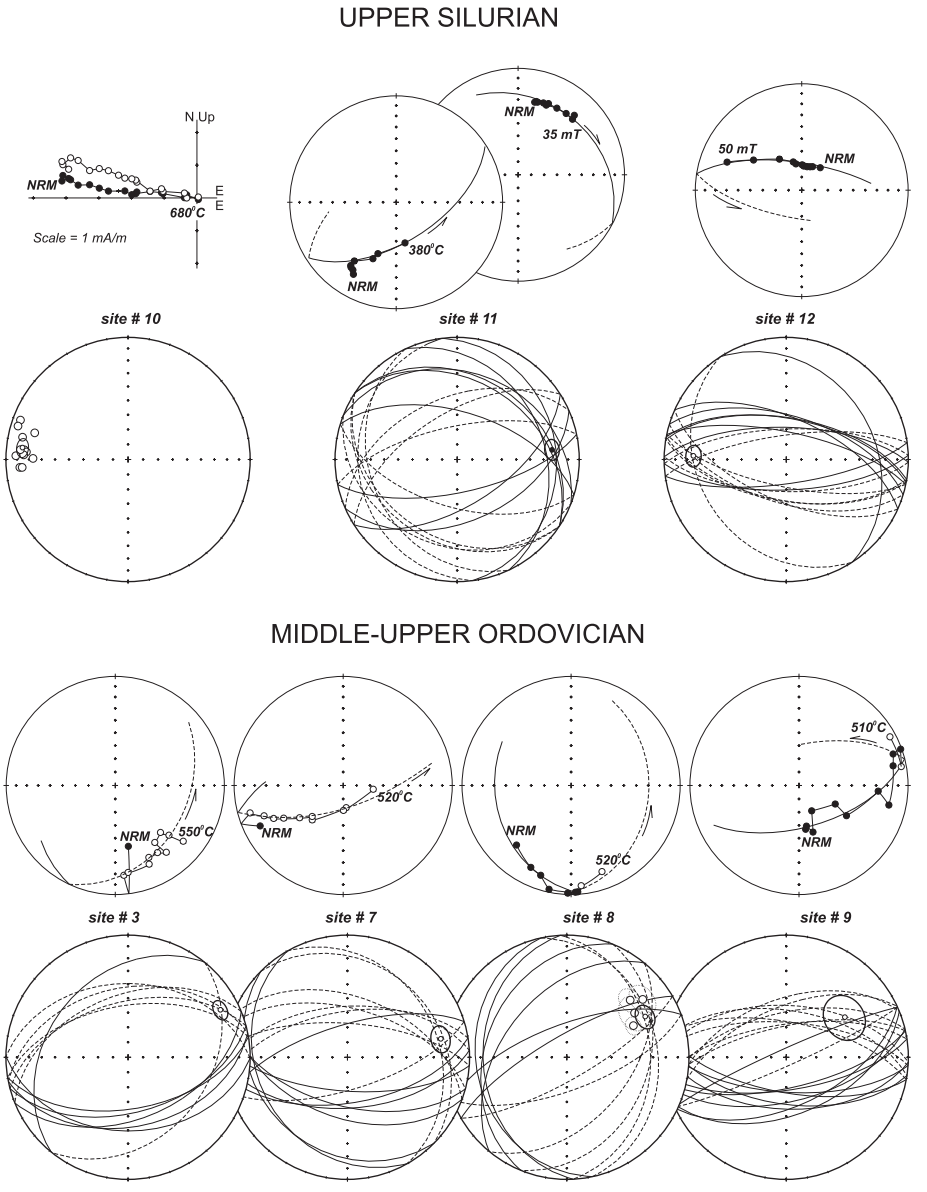


Fig. 4. Typical sample behaviour during stepwise thermal and/or AF demagnetization (orthogonal plots for principal component analysis and stereoplots for great circles method) and resulting stereoplots of each site for Upper Silurian (sites 10, 11, 12) and Ordovician (sites 3, 7, 8, 9). All plots are represented in stratigraphic system. Legend: (i) For orthogonal plots: solid (open) dots represent projections of vector endpoint on the horizontal (vertical) plane. (ii) For stereoplots: solid (open) dots and solid (dashed) lines represent vector endpoint projection and great circle path on upper (lower) hemisphere, respectively; dots surrounding by ovals represent mean site direction with corresponding confidence limit.

temperature linear segments are not observed, and at 310–340 °C, the NRM intensity approached the instrumental noise level. The low unblocking temperatures indicate that the most probable carrier of magnetization is titanomaghemite. Those results are supported by saturation isothermal (SIRM) demagnetization experiments. During heating, titanomaghemite transforms into magnetite and hematite (Dunlop and Ödzemir, 1997), there is a sharp increase SIRM (more than 50 times) after the first

heating and completes decay of this magnetization near unblocking temperatures of magnetite. MDF values are very low (maximum 4 mT), and coercivity components <30 mT are present, but with larger scatter than those identified with thermal demagnetization. At higher AF fields, directions are distributed along great circles that converge toward ChRCs identified from the sandstones (site 10). The great-circle intersection method allows us to isolate a site mean direction, which is similar to those from the sandstones in stratigraphic coordinates (Fig. 4).

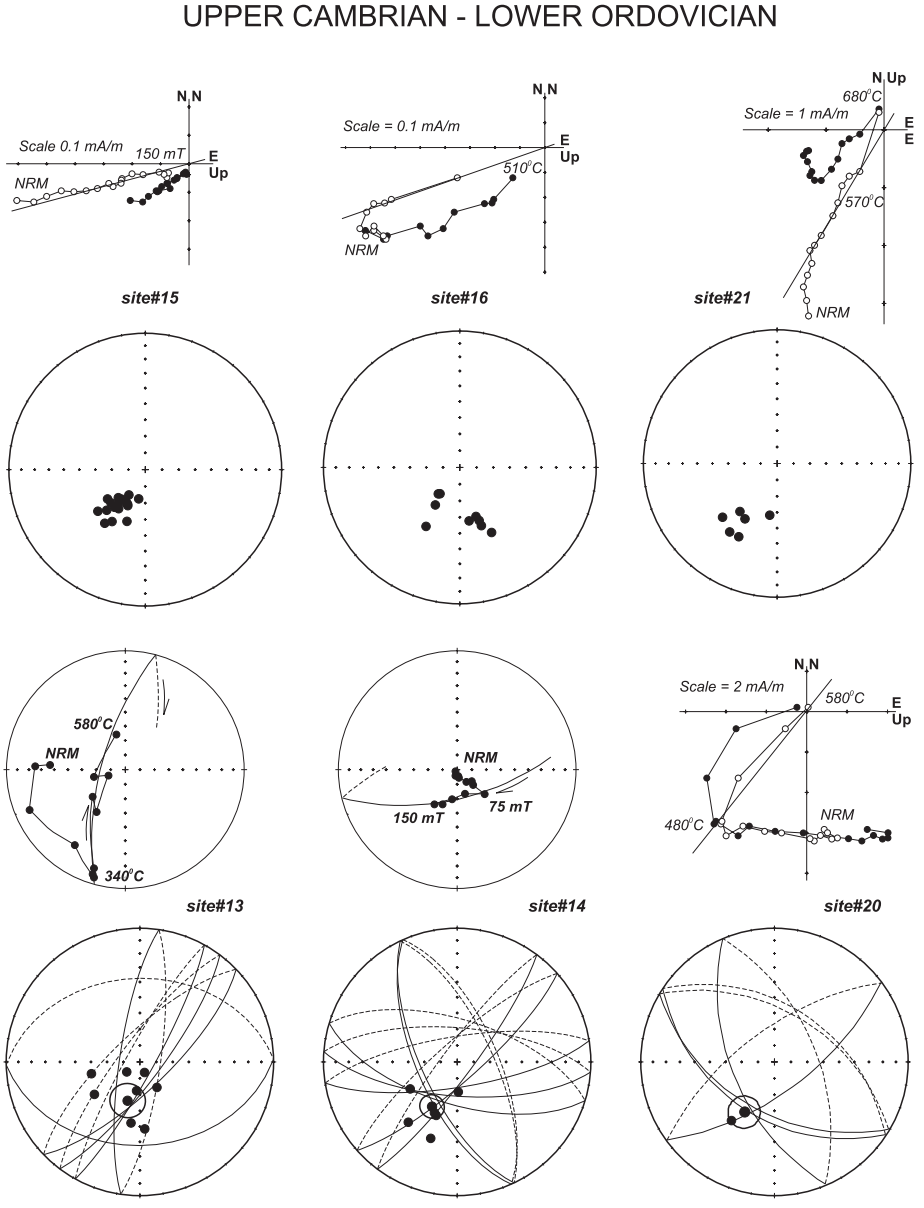


Fig. 5. Typical sample behaviour during stepwise thermal and/or AF demagnetization (orthogonal plots for principal component analysis and stereoplots for great circles method) and resulting stereoplots of each site for Upper Cambrian (sites 15, 16, 21, 13, 14, 20). All plots are represented in stratigraphic system. Legend is the same as for Fig. 4.

Table 1
Comparison of mean site directions obtained using different methods (site 14)

Method	N	Mean direction				Angle between mean directions (γ/γ_c)		
		D (°)	I (°)	k	α_{95} (°)	PCA	COMB	GC
PCA	6d	208.9	54.5	25.5	13.5	–	4.0/14.1	10.1/14.2
COMB	6d+7c	209.4	58.5	34.9	7.3	4.0/14.1	–	6.2/8.6
GC	7c	207.6	64.6	213.4	5.0	10.1/14.2	6.2/8.6	–

Method: PSA–principal component analysis of individual direction (Fisher, 1953, statistics); COMB–combined analysis of individual directions and great circles (McFadden and McElhinny, 1988); GC–great circle method (Halls, 1976). *N*–number of samples used in statistics (d–individual directions, c–great circles); *D*–mean declination; *I*–mean inclination, *k*–precision parameter; α_{95}–95% error parameter; γ–angle between two directions; γ_c–critical value of the angle.

4.2. Middle–Upper Ordovician

NRM intensities for the Ordovician samples average about 1 mA/m, whereas the magnetic susceptibility varies from 10^{-6} to 10^{-5} SI units. Thermal demagnetization reveals that the NRM is composite and that at least two components are present, in addition to a viscous magnetization component that is demagnetized at low temperatures or low AF fields. For similar reasons as advocated above for sites 11 and 12 (i.e., overlapping component spectra, thermochemical changes induced during heating, NRM intensity dropping below the instrument noise level at high temperatures or AF fields), high unblocking components were normally not isolated or identified during demagnetization experiments. An exception was a few samples from site 8, which revealed northeast declinations and negative inclinations (stratigraphic coordinates). However, the remaining demagnetization data are distributed along great-circle trajectories and it was possible to estimate site-mean high unblocking components by calculating the intersection of these great circles (Fig. 4).

4.3. Upper Cambrian–Lower Ordovician

NRM intensities are somewhat higher than the Upper Ordovician and Upper Silurian samples, with the NRM intensity as high as 10 mA/m, whereas the magnetic susceptibility varies from 10^{-5} to 10^{-4} SI units. The volcanic samples from sites 15, 16 and sandstone samples from site 21 demonstrate relatively high stability to demagnetization with respect to temperature and AF treatments (Fig. 5). For the volcanic rocks, the remanence is carried by high-coercivity (fine-grained) magnetite, whereas hematite dominates in the sandstone samples. Both minerals are evident from thermal demagnetization spectra, SIRM demagnetization and reflected light microscopy. ChRCs, estimated from AF and thermal treatment, are similar, yielding SSW declinations and positive inclinations (stratigraphic coordinates) (Fig. 5). Low-temperature/coactive components do not demonstrate any regular direction.

Initial magnetization of sandstones from sites 13, 14 and volcanic rocks from site 20 was party destroyed by later remagnetization. Blocking temperatures and coercive spectra for different components essentially overlap during demagnetization. The vector endpoint follows a great circle arc, generally towards the ChRC means of sites 15–16 and 21. Although orthogonal plots allow isolation of the stable component in some samples, the confidence cone of this direction is relatively large. This is connected with incomplete separation of components during demagnetization and low values of remanent magnetization (relative to the noise of the measuring instrument). To estimate mean directions for each of the investigated objects, the method of convergence of great circles was used. Comparison of means obtained using principal component analysis and the combined analysis of components and circles with those from only great circle method do not demonstrate any shallowing of inclination (Table 1).

5. Interpretation of paleomagnetic data

Paleomagnetic site mean directions, paleolatitudes and pole positions are summarized in Fig. 6 and Table 2. In situ (geographic coordinate) site means are

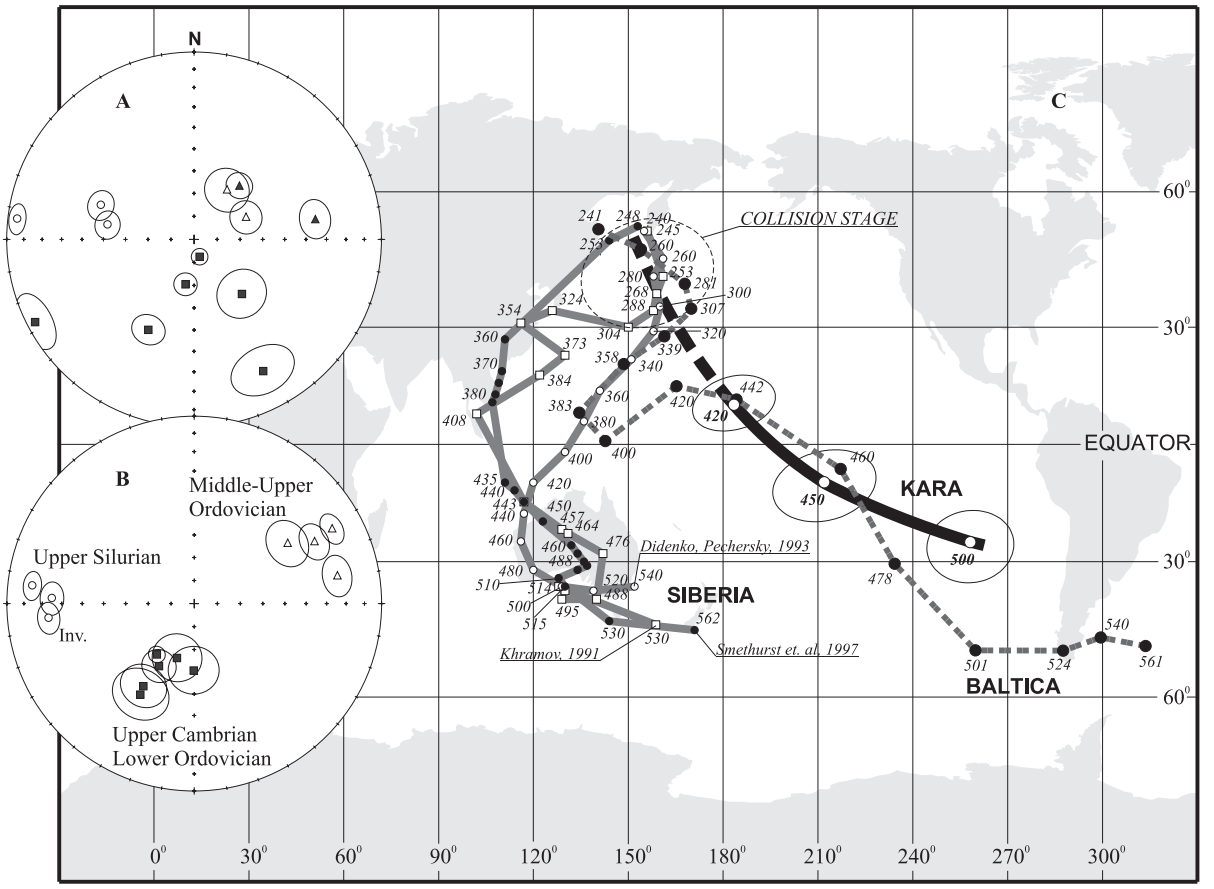


Fig. 6. Paleomagnetic site mean directions in situ (a) and tilt corrected (b). Apparent polar wander path (APWP) for Kara and its comparison with Siberia and Baltica APWPs (c).

scattered, but upon bedding correction (stratigraphic coordinates), the Upper Silurian site means group with shallow negative inclinations due west, the Middle–Upper Ordovician site means show intermediate negative inclinations with ENE declinations, and the Upper Cambrian–Lower Ordovician site means cluster with southerly declinations and intermediate to steep positive inclinations.

The three Silurian site means tends cluster upon unfolding, with an increase of precision parameter k from 10.2 in situ to 114.5 after tilt correction. The ratio of precision parameters ($ks/kg=11.2$) is higher than the critical value at the 95% confidence level (6.55), and the fold test (McElhinny, 1964) is therefore positive. The fold test of McFadden (1990) gives the same result: in geographic coordinate $\xi_1=2.772$, in stratigraphic coordinate $\xi_1=1.890$, critical value of the test statistic $\xi=2.662$ at the 99% confidence level indicative of positive fold test. The most probable time of folding of these strata is Late Carboniferous–Early Permian (Zonenshain et al., 1990; Bogdanov et al., 1998). The magnetization is therefore pre-Permian in age and is inferred to be close to the age of formation of these sediments, i.e., a Wenlock–Ludlow age (430–415 Ma).

Stepwise unfolding of the four Middle–Late Ordovician site means shows that the maximum of precision parameter k is achieved at complete unfolding. The k parameter is significant higher in stratigraphic coordinates ($ks=38.8$) than the geographic coordinates ($kg=1.6$) and the fold test (McFadden, 1990) demonstrates $\xi_1=3.884$ in geographic coordinates and $\xi_1=1.136$ in stratigraphic coordinates while critical value of the test statistic is

Table 2
Paleomagnetic data from the Paleozoic section of October Revolution Island

Rock unit	*n*	Geographic coordinates				Stratigraphic coordinates			
		D (°)	*I* (°)	α_{95} (°)	*k*	D (°)	*I* (°)	α_{95} (°)	*k*
Upper Silurian (430–415 Ma)									
Ust'spokoininskaya Fm., sandstones, site 10	12d	276.5	−5.6	5.1	73.7	276.2	−14.1	5.1	73.7
Samoilovichskaya Fm., limestones, site 11*	9c	279.4	−51.3	5.8	101.7	264.2	−22.9	5.8	102.5
Samoilovichskaya Fm., marls, site 12	10c	290.0	−46.0	6.1	79.9	272.0	−24.9	6.1	79.9
Site means (sampling coordinates 79.6°N, 96.3°E)	3	281.3	−34.7	40.9	10.2	270.8	−21.0	11.6	114.5
	Pole: 10.5°N, 183.5°E d*p*/d*m* = 6.4/12.2 Plat: 10.9 ± 8.8								
Middle–Upper Ordovician (465–440 Ma)									
Ushakovskaya Fm., limestones, site 3	7c	40.1	59.2	5.7	165.3	61.6	−17.4	5.7	164.0
Stroininskaya Fm., sandstones, site 7	8c	80.5	34.7	7.6	73.1	79.1	−23.2	7.6	73.2
Stroininskaya Fm., sandstones, site 8	4d+6c	67.0	−65.4	7.1	50.9	62.8	−28.6	7.1	50.7
Stroininskaya Fm., sandstones, site 9	10c	33.7	−64.1	9.6	31.0	57.2	−40.4	9.6	31.0
Site means (sampling coordinates 79.6°N, 96.3°E)	4	60.8	−9.8	122.0	1.6	66.4	−27.2	14.9	38.8
	Pole: 10.1°S, 212.0°E d*p*/d*m* = 8.8/16.2 Plat: −14.4 ± 12.0								
Upper Cambrian–Lower Ordovician (510–480 Ma)									
Kruzhilikhinskaya Fm., effusives, site 15	17d	162.3	82.1	3.6	98.2	216.3	62.4	3.7	96.3
Kruzhilikhinskaya Fm., effusives, site 16	9d	138.8	58.2	9.9	27.8	180.5	60.7	10.7	23.9
Kurchavinskaya Fm., sandstones, site 13	8d+5c	205.2	58.2	10.6	16.6	197.6	65.1	10.6	16.6
Kurchavinskaya Fm., sandstones, site 14	6d+7c	206.8	66.4	7.3	34.8	209.4	58.5	7.3	34.9
$\in_3$-O_1 undivided, effusive, site 20	1d+4c	155.3	27.9	11.7	67.7	210.6	42.9	11.6	68.0
$\in_3$-O_1 undivided, sandstones, site 21	6d	242.5	4.9	9.7	48.9	211.6	47.0	9.7	48.9
Site means (sampling coordinates 78.5°N, 98.2°E)	6	191.4	57.2	34.4	4.7	205.7	56.8	9.4	51.4
	Pole: 26.9°S, 255.5°E d*p*/d*m* = 9.9/13.6 PLat: −37.4 ± 11.6								

n–number of samples (sites for mean direction); d–number of individual directions, c–number of circles used; *D*, *I*, α_{95}, *k*–remanence declination, inclination, 95% error parameter, the precision parameter by Fisher (1953); Pole–paleomagnetic pole, d*p*/d*m*–semiaxes of the cone of 95% confidence about the pole; Plat–paleolatitudal position for block, *for statistics the inverted direction in stratigraphic (not in geographic) coordinates is used.

$\xi = 3.180$ at the 99% confidence level. Using the method of Watson and Enkin (1993), the optimal concentration is found at 113.3 ± 5.1 (95% confidence interval) close enough to 100%, so the remanence can be considered as pretilting. We therefore assume the time of remanence acquisition corresponds to the time of deposition, i.e., Middle–Late Ordovician (between 460 and 440 Ma).

Mean directions of the Late Cambrian–Early Ordovician sites (in stratigraphic coordinates) coincide within the confidence limits (Table 1). The precision parameter *k* increases upon untilting from *kg* = 4.7 to *ks* = 51.4. The McFadden's (1990) fold test parameters are $\xi_1 = 4.522$ in geographic coordinates and $\xi_1 = 1.113$ in stratigraphic coordinates. Critical value of the test statistic $\xi = 3.919$ at the 99% confidence level indicative of positive fold test. The fold test of Watson and Enkin (1993) gives an optimum degree of untilting at 104% (with 95% confidence limits at 95.3% and 112.4%) and indicate that remanence is indeed pretilting. We infer that the magnetizations are primary and assign a mean age of 500 Ma (510–480 Ma).

The positive fold-test, the indication of dual polarity in the Upper Silurian data set, and the different mean directions for all the three sampled time periods (ca. 420 Ma, 450 Ma and 500 Ma) point to the near primary character of ChRCs obtained. These results therefore allow the construction of an APW path for the Kara microcontinent for the first time and a comparison of this path with available paths for Siberia (Khramov, 1991; Smethurst et al., 1998) and Baltica (Torsvik et al., 1996; Torsvik and Rehnström, 2001).

We suggest that the Kara poles plotted in Fig. 6 are north poles and therefore position the Kara micro-

continent in the southern hemisphere during the Early Paleozoic. The APW path for Kara follows a smooth NW trend, starting at around 25°S (Late Cambrian), and ending up at ca. 10°N in the West Pacific (Late Silurian). This Early Paleozoic APW path differs markedly from that for Siberia, but shows some similarities with that of Baltica. Late Paleozoic (Devonian to Permian) poles from Kara have not yet been obtained. However, we know from the geological record (i.e., in the South Taimyr Domain) that the initial collision of Kara with Siberia took place in Late Carboniferous to Permian time (Vernikovsky, 1996).

6. Drift history and geodynamics

The APW path (Fig. 6) allows us to calculate latitudinal velocities and angular rotations for the Kara microcontinent (Fig. 7). From the end of the Cambrian to the end of the Silurian, the Kara microcontinent underwent systematic northward drift from intermediate southerly latitudes (ca. 40°S) to low subtropical latitudes (ca. 10°N). The latitudinal velocity (minimum plate velocity since paleolongitude is unknown) apparently increased from 5 to 10 cm/year during the Early Paleozoic, while angular rotations (<1°/Ma) led to a net counter-clockwise rotation. The estimates of

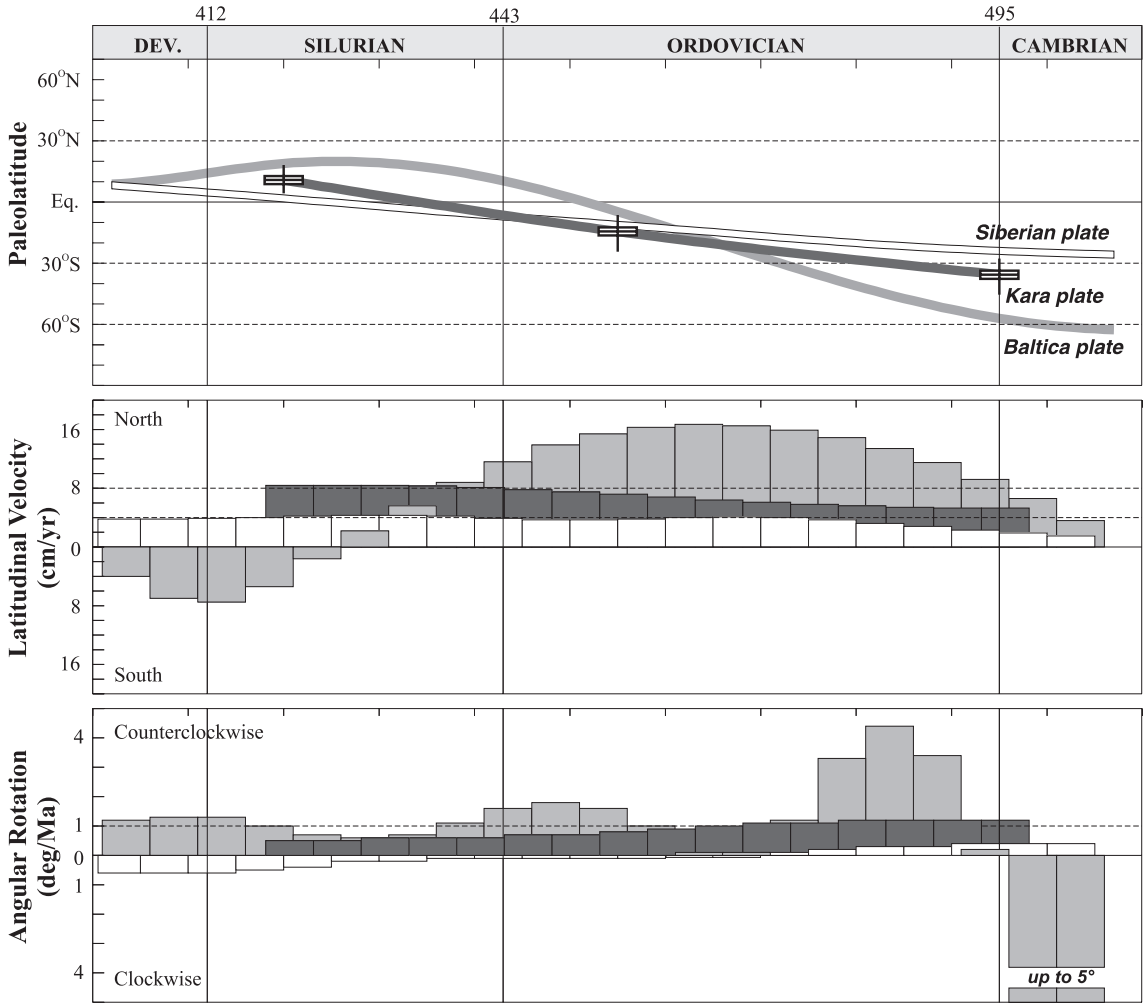


Fig. 7. Paleolatitudinal (top diagram) drift history and latitudinal (middle diagram) and angular (lower diagram) velocity of Kara over geological time (reference location is 79°N 97°E) and comparison with Baltica and Siberia drift and rotation history.

both the latitudinal velocities and rotation rate are compatible with measured values for present-day small continental plates.

As shown in Fig. 6, there are three published APW paths for Siberia during the Paleozoic, by Khramov (1991), Didenko and Pechersky (1993), and Smethurst et al. (1998), respectively. However, despite some differences in detail, the general character of the three paths is quite similar. For example, based on the paleomagnetic results of Didenko and Pechersky (1993), the latitudinal drift velocity of Siberia did not exceed about 5 cm/year, a result comparable to that determined by Smethurst et al. (1998). However, the latter authors proposed that there was a sharp bend in the APWP associated with abrupt changes in plate velocity at about 510–480 Ma which is not recorded in the results of the other authors. According to the paleomagnetic data, Siberia was gradually rotating clockwise during the Paleozoic, so that by Triassic time, its southern margin faced north. This can be reconciled with motion along a great circle arc (Khramov and Rodionov, 1980) and implies the presence of large-scale strike–slip faults.

The initial paleolatitudinal convergence of Siberia and Kara occurred approximately during the Middle–Late Ordovician, indicating that the two plates were in the same geographic region by that time. However, their strongly divergent APW trajectories provide unequivocal evidence for their separate existence at that time (Fig. 6). Furthermore, paleomagnetic data require a significant relative rotation between Siberia and Kara before the final collision, implying that they were separated by large strike–slip fault zones.

Models describing the Paleozoic tectonic evolution of Siberia, especially the southwestern part of the Siberian platform (e.g., Berzin and Dobretsov, 1994; Kungurtsev et al., 2001; Gibsher, 2001) emphasize the major role of strike–slip tectonics, which is inferred to have resulted in the clockwise rotation of Siberia, and it is logical to assume a similar style of shear kinematics for the northern margin of the Siberian craton (i.e., the Taimyr–Severnaya Zemlya fold belt). Similarly, modern structures of this area are connected to sinistral large-scale strike–slip faults.

The kinematics of Baltica differs significantly to those determined for Kara. According to Torsvik et al. (1996), the APW path for Baltica during the Ordovician to Silurian indicates a northward drift from moderate southerly latitudes (about 50°S) to subtropical latitudes (up to ~20°N). Latitudinal velocity was higher than for both Siberia and Kara, reaching 8–10 cm/year (Torsvik et al., 1996). In the Late Silurian, collision between Baltica and Laurentia caused a change in drift direction (Torsvik et al., 1996), but by Devonian time, the northerly sense of horizontal motion was restored. At the beginning of the Paleozoic, the northwestern margin of Baltica (the Caledonian suture) was rotated towards the Taimyr margin of Siberia. Angular counterclockwise rotation of Baltica from the Cambrian until the middle Devonian was connected to negligible clockwise rotation of Siberia. The average rotational velocity of Baltica's motion exceeded 1°/Myr, i.e., was appreciably faster than rotation of Kara and Siberia. The comparatively fast counter-clockwise rotation of Baltica in the Early Paleozoic correlates well with expansion of Iapetus Ocean and with the postulated occurrence of large-scale transform zones between the approaching Baltica and Siberia plates (Torsvik et al., 1996). This tectonic model implies that Baltica was rotated almost 180° during the Early Paleozoic (Trench and Torsvik, 1992; Torsvik et al., 1996).

In order to compare the relative motions of Baltica and Kara, the velocity characteristics of Baltica are given in coordinates of the Kara microcontinent (Fig. 7). The differences in angular and latitudinal motions of the plates are evident, although the direction of latitudinal drift and rotation are uniform. If the Kara terrane had been a part of Baltica or Siberia during the Early Paleozoic, then velocity parameters for their motions should coincide. The intersection of paleolatitude curves for Baltica, Kara, and Siberia in the middle Ordovician (Fig. 7) testifies a close latitudal position of those blocks, but does not provide evidence for their collision at that time.

7. Paleogeography and a new collision model for Siberia and Kara

The paleogeographic reconstructions in Fig. 8 comprise the results of recent integrated paleomagnetic and paleogeographic investigations that were

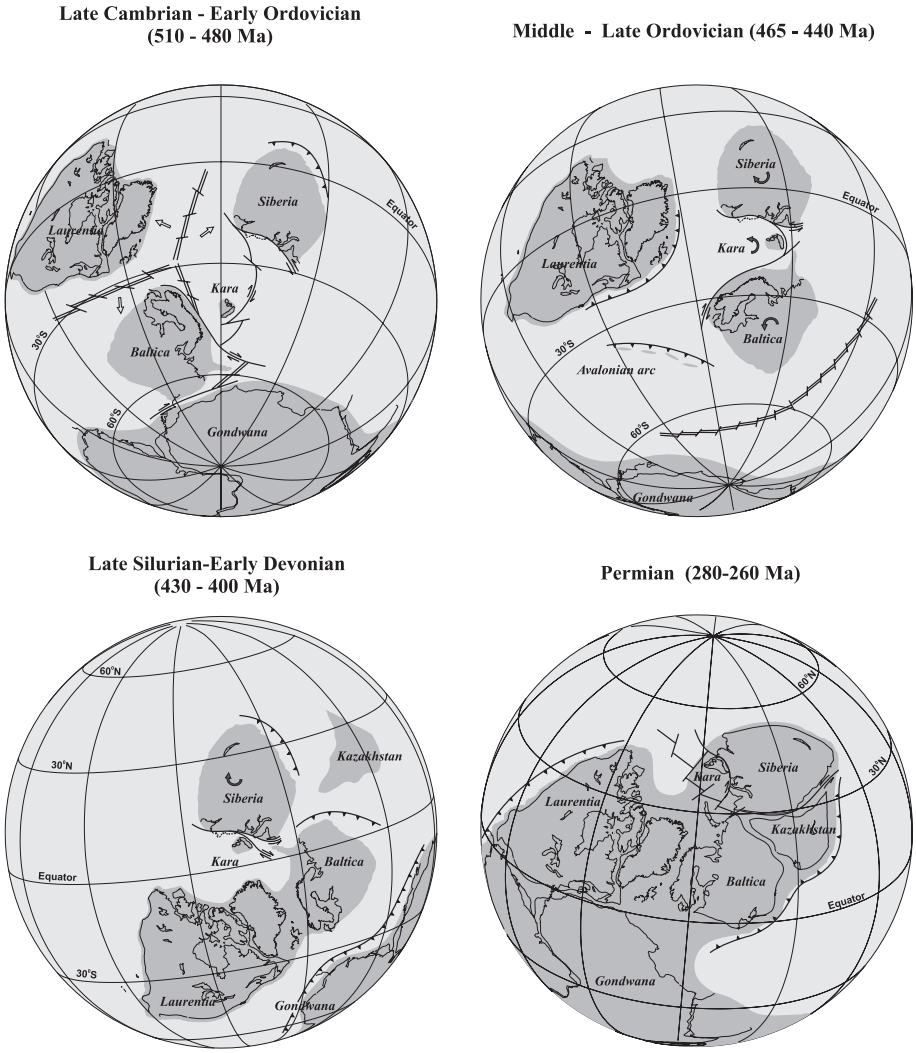


Fig. 8. Paleogeographic scheme of drift history of the Kara microcontinent and relation to other continents during Paleozoic.

assembled for a series of Phanerozoic paleogeographic maps and global reconstructions (Khramov and Ustritsky, 1990; Scotese and McKerrow, 1990; Torsvik et al., 1996; Scotese, 1997; Torsvik, 1998; Cocks, 2000). The orientations and latitudinal positions of Laurentia are taken from the reconstruction of Torsvik et al. (1996), those for the Gondwana plates from Van der Voo (1988), and for Baltica from Torsvik et al. (1996), Torsvik and Rehnström (2001) and Didenko et al. (2001). The position of Siberia follows the interpretations of Didenko and Pechersky (1993) and Smethurst et al. (1998).

7.1. Late Cambrian to early Ordovician

This was a time of active spreading between Laurentia on the one hand, and both Baltica and Siberia on the other (Pickering and Smith, 1995). Rift formation and growth of the Iapetus Ocean followed breakup of the Rodinia supercontinent in the Neoproterozoic. Supercontinent breakup should be marked by rifting on continental shelf margins and separation of plates into independent terrains, including small micro-plates such as Kara. The arrangement of spreading zones that gave rise to the formation of Iapetus also resulted in the counter-

clockwise rotation of Baltica and the formation of an extended transform zone on its margins. At the Cambrian–Ordovician boundary, the margins of Siberia and Baltica were both arenas of accretional–collisional events. For instance, on the southwestern margin of Siberia, accretion of a Late Neoproterozoic to Cambrian island arc system to the continent resulted in development of the central part of Altay-Sayan fold belt (Berzin and Kungurtsev, 1996), which is also characterized by large transform shear zones (Gibsher, 2001; Kungurtsev et al., 2001). Subsequently, Siberia rotated clockwise. The northern margin of Siberia, after its collision with the Central Taimyr accretionary terrane in the Late Neoproterozoic, began to develop as a passive continental margin, with a platformal sedimentary environment (Pogrebitsky, 1971; Vernikovsky, 1996).

In the northwestern part of Baltica, the first phase of a collision (Finnmarkian stage) is thought to have occurred at the Cambrian–Ordovician boundary and is also connected to the accretion of the island arcs (Stephens and Gee, 1989; Sturt and Roberts, 1991). After closure of the Paleo-Uralian Ocean and Neoproterozoic orogenesis, the eastern (Uralian) margin of Baltica underwent rifting in Cambrian–Ordovician time (Zonenshain et al., 1990). The age of the base of the sedimentary cover that unconformably overlaps Neoproterozoic folded complexes in the western Urals decreases from the north to the south (in modern coordinates), being Middle Cambrian in age west of Novaya Zemlya, Upper Cambrian in the Northern and Polar Urals, and Ordovician in the southern Urals (Zonenshain et al., 1990). This age difference is related to the progress of rifting during the north-to-south breakup of the Uralian margin of Baltica. The basal part of the section is generally formed by sedimentary deposits resulting from denudation of the Baltica basement, but locally the rift succession is floored by bimodal volcanic complexes indicating that magmatism was associated with the rifting process (Zonenshain et al., 1990). The breakup was probably preceded by brittle deformation connected with dextral strike–slip faulting (Torsvik and Rehnström, 2001). Overall, the rifting and opening of the Uralian Ocean is compatible with counter-clockwise rotation of Baltica during the Early Paleozoic.

A rifting stage is also manifested in the stratigraphy of the Kara microcontinent, where Early Ordovician rift sediments, volcanic rocks and comagmatic intrusives of alkaline and subalkaline composition are present on October Revolution and Komsomolets islands. Rhyolite porphyry, felsite, porphyritic granite, ignimbrite and other volcanic rocks of mafic to felsic composition occurring in the Sverdlov Cape area, southern October Revolution Island, are also attributed to this complex (Proskurnin, 1995). U/Pb dating of zircons from an analogous volcanic complex to the north of Sverdlov Cape has yielded an age of 492 ± 7 Ma (Lorenz and Gee, 2000).

Thus, this brief review of the geodynamics of Baltica and Siberia during the Cambrian to Early Ordovician allows the interpretation that the northward drift of the Kara microcontinent was probably caused by the existence of an extensive system of transform faults between Siberia and Baltica in the Early Paleozoic. Formation of the fault system occurred during rifting of these continental plates, with the strike–slip faults resulting from the subsequent relative rotation of the continents. This model provides the simplest explanation for the kinematics of the Kara microcontinent and its collision with Siberia.

7.2. Middle–Late Ordovician

During the Ordovician, Baltica underwent counter-clockwise rotation due to rifting, whereas Siberia rotated in the opposite direction. At the same time, the Kara microcontinent drifted northward along a system of dextral strike–slip faults with a simultaneous counter-clockwise rotation. Throughout the Middle to Late Ordovician, all three continents (Kara, Baltica and Siberia) were displaced towards the north and the Kara entered the tropics, causing a gradual change in sedimentary environment and a concomitant change in deposition from siliciclastics to carbonate. Up to this time, the Uralian part of Baltica was a stable passive continental margin, which indicates the presence of a mature ocean to the east (in modern coordinates) (Didenko et al., 2001). The geodynamic setting of the inner part of the Kara microcontinent did not undergo significant change and continued to develop as a platformal regime accumulating fine-grained sandstone, dolostone, clayey limestone and marl during the Late Ordovician. The width of the ocean basins separating the Kara microcontinent from

Siberia and Baltica was not large at this time and did not represent an obstacle for faunal migration. The Late Ordovician paleolatitudes for Siberia and Kara, on the one hand, and Kara and Baltica, on the other, are practically indistinguishable, providing evidence for the inference of spatial proximity of these two paleobiogeographic provinces. Therefore, the presence in strata on Severnaya Zemlya of Ordovician fauna that are characteristic of both the Siberian and east European paleobiogeographic provinces (Bogolepova et al., 2001) is quite reasonable.

7.3. Late Silurian

By the end of the Ordovician and the beginning of the Silurian, the extensional tectonic setting was replaced by a compressional regime, and oceans separating the three continents began to close. Island arc complexes of the eastern Urals (Zonenshain et al., 1990; Puchkov, 1997; Didenko et al., 2001) provide evidence of subduction at this time. As a result of the tectonic events caused by closure of the ocean between Laurentia and Baltica, the Caledonian collision belt was formed at the end of Silurian (Torsvik et al., 1996). The paleogeographic position of the continents at this time can be characterized as a subequatorial on the basis of paleomagnetic data (Torsvik et al., 1996; Smethurst et al., 1998; Didenko and Pechersky, 1993) and carbonate deposits, including "Bahamian" type reefs.

By the end of the Silurian, the Kara microcontinent was proximal to the margin of Siberia (their paleolatitudes coincide), but the Late Silurian pole position of Kara differs from that of Siberia. However, Late Paleozoic subduction complexes have not been found on Taimyr, and thus, early collisional events may have been connected with transform or strike–slip faults and the discrepancy of paleomagnetic pole positions resulted from relative rotations about local pivots. Nevertheless, the presence of oceanic crust fragments between Siberia and Kara must not be ruled out.

Large-scale transform strike–slip shear zones, which brought Kara into juxtaposition with Siberia during the Early Paleozoic, are inferred to have finally resulted in its collision with Siberia and the subsequent formation of the nappe-overthrust structures of the Taimyr–Severnaya Zemlya fold belt. The orientation of the main thrusts, including the Main Taimyr suture, may also imply the opposed senses of rotation of Siberia and Kara during the collision stage, as postulated in the kinematic model. Nappes and thrust faults in eastern Taimyr are oriented NE, whereas farther west they gradually change to an E–W orientation (Fig. 1) and the angle of rotation increases regularly from the periphery to the central parts of North Taimyr Domain. This imbricated structure could have been generated under horizontal compression and simultaneous counter-clockwise rotation of Kara.

A similar inference, based on analysis of paleobiogeographic data and several reconstructions of the region for Late Devonian–Permian times has been already expressed (Khramov and Ustritsky, 1990). According to these authors, the positions of Siberia and Kara were close to those proposed in our model. Geologic evidence indicates that final collision of the continents occurred in the Late Carboniferous to Permian (Vernikovsky et al., 1995, 1998b; Vernikovsky, 1996, 1997; Bogdanov et al., 1998).

If it is assumed that the kinematic model described above continued during the Late Paleozoic, then the convergence of APWP curves for Siberia and Kara would occur in the interval of 300–250 Ma, corresponding to the known Late Carboniferous–Permian time of collision. Thus, the collision age derived from the proposed model is in good agreement with the available geological data.

8. Conclusions

(1) The northern part of Taimyr Peninsula and adjacent regions of the Kara Sea and Severnaya Zemlya Archipelago are underlain by the Kara microcontinent, which was a separate tectonic block during the Paleozoic.
(2) During the Early Paleozoic, the Kara microcontinent was located between Baltica and Siberia and drifted northwards from moderate latitudes (about 40°S) in the southern hemisphere to subequatorial latitudes (about 10°N) in the northern hemisphere with a simultaneous counter-clockwise rotation of more than 65°.
(3) These motions of the Kara microcontinent are interpreted to have been related to large-scale transform strike–slip fault zones.

(4) The approach and subsequent collision of the Kara microcontinent with Siberia in Late Carboniferous to Permian time occurred as a result of strike–slip tectonics. Deformation of the Paleozoic margins of Siberia and Kara occurred as a result of differential rotations of these continental blocks.

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